SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
CHENNAI – 602105

CSA0501-DATABASE MANAGEMENT SYSTEMS

LAB EXPERIMENTS **DATE: 20/07/2026**

EXPERIMENT 7 -- SELECT with various clause – GROUP BY, HAVING, ORDER BY

AIM:

To retrieve and organize records from the **student_cgpa** table using the **GROUP BY**, **HAVING**, and **ORDER BY** clauses in MySQL Workbench.

PROCEDURE:

1. Open MySQL Workbench and connect to the MySQL server.
2. Select the required database using the USE command.
3. Create the student_cgpa table and insert the required records.
4. Display the table contents using the SELECT statement.
5. Use the **GROUP BY** clause to group records based on the department.
6. Apply the **HAVING** clause to filter grouped results.
7. Use the **ORDER BY** clause to sort records in ascending and descending order.
8. Execute each query and verify the output in the Result Grid.
9. Observe the grouped, filtered, and sorted results.
10. Conclude the experiment after successful execution.

PROGRAM:

1. CREATE TABLE

```
CREATE TABLE student_cgpa (
reg_no VARCHAR(15) PRIMARY KEY,
student_name VARCHAR(50) NOT NULL,
department VARCHAR(20),
no_of_subs_pass INT,
no_of_subs_fail INT,
cgpa DECIMAL(3,2)
);
```

Then Insert Values:

```
INSERT INTO student_cgpa
(reg_no, student_name, department, no_of_subs_pass, no_of_subs_fail, cgpa)
```

```
VALUES
('192411123','Deepthi','CSE',22,2,8.71),
('192512121','Varun','IT',27,1,9.23),
('192312117','Aadhya','ECE',25,3,8.97),
('192411227','Dimple','CSE',31,0,9.37),
 ('192820896','Mohan','EEE',18,9,6.48),
 ('192411122','Yogitha','CSE',31,1,9.01),
 ( '192516871','Sadhya','IT',25,5,7.22),
 ('192111101','Rahul','MECH',30,12,5.14),
  ('192411010','Alekhya','CSE',35,2,9.01),
 ('192511135','Rahul','CIVIL',37,0,9.72);

 SELECT * FROM student_cgpa;
```

OUTPUT:

Result Grid | Filter Rows: | Edit: | Export/Import:

reg_no	student_name	department	no_of_subs_pass	no_of_subs_fail	cgpa
192111101	Rahul	MECH	30	12	5.14
192312117	Aadhya	ECE	25	3	8.97
192411010	Alekhya	CSE	35	2	9.01
192411122	Yogitha	CSE	31	1	9.01
192411123	Deepthi	CSE	22	2	8.71
192411227	Dimple	CSE	31	0	9.37
192511135	Rahul	CIVIL	37	0	9.72
192512121	Varun	IT	27	1	9.23
192516871	Sadhya	IT	25	5	7.22
192820896	Mohan	EEE	18	9	6.48

Result Grid | Form Editor | Field Types

udent_cgpa 1 × | Apply | Revert | Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	22:41:47	CREATE TABLE student_cgpa (reg_no VARCHAR(15) PRIMARY KEY, student_name VARCHAR(50) NO...	0 row(s) affected	0.031 sec
2	22:44:31	INSERT INTO student_cgpa (reg_no, student_name, department, no_of_subs_pass, no_of_subs_fail, cgpa) VA...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
3	22:46:18	SELECT * FROM student_cgpa LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec

GROUP BY

1. Count the number of students in each department.

```
SELECT department, COUNT(*) AS Number_of_Students
FROM student_cgpa
GROUP BY department;
```

OUTPUT:

Result Grid | Filter Rows:

department	Number_of_Students
MECH	1
ECE	1
CSE	4
CIVIL	1
IT	2
EEE	1

2. Find the average CGPA of each department.

```
SELECT department, AVG(cgpa) AS Average_CGPA
FROM student_cgpa
GROUP BY department;
```

OUTPUT:

Result Grid | Filter Rows:

department	Average_CGPA
MECH	5.140000
ECE	8.970000
CSE	9.025000
CIVIL	9.720000
IT	8.225000
EEE	6.480000

3. Find the maximum CGPA in each department.

```
SELECT department, MAX(cgpa) AS Maximum_CGPA
FROM student_cgpa
GROUP BY department;
```

OUTPUT:

Result Grid | Filter Rows:

department	Maximum_CGPA
MECH	5.14
ECE	8.97
CSE	9.37
CIVIL	9.72
IT	9.23
EEE	6.48

4. Find the minimum CGPA in each department.

```
SELECT department, MIN(cgpa) AS Minimum_CGPA
FROM student_cgpa
GROUP BY department;
```

OUTPUT:

department	Minimum_CGPA
MECH	5.14
ECE	8.97
CSE	8.71
CIVIL	9.72
IT	7.22
EEE	6.48

5. Find the total number of passed subjects in each department.

```
SELECT department, SUM(no_of_subs_pass) AS Total_Passed_Subjects
FROM student_cgpa
GROUP BY department;
```

OUTPUT:

department	Total_Passed_Subjects
MECH	30
ECE	25
CSE	119
CIVIL	37
IT	52
EEE	18

HAVING

6. Display departments having more than one student.

```
SELECT department, COUNT(*) AS Number_of_Students
FROM student_cgpa
GROUP BY department
HAVING COUNT(*) > 1;
```

OUTPUT:

department	Number_of_Students
CSE	4
IT	2

7. Display departments whose average CGPA is greater than 8.50.

```
SELECT department, AVG(cgpa) AS Average_CGPA
FROM student_cgpa
GROUP BY department
HAVING AVG(cgpa) > 8.50;
```

OUTPUT:

113

department	Average_CGPA
ECE	8.970000
CSE	9.025000
CIVIL	9.720000

8. Display departments whose total passed subjects are greater than 50.

```
SELECT department, SUM(no_of_subs_pass) AS Total_Passed_Subjects
FROM student_cgpa
GROUP BY department
HAVING SUM(no_of_subs_pass) > 50;
```

OUTPUT:

department	Total_Passed_Subjects
CSE	119
IT	52

ORDER BY

9. Display all students in ascending order of Registration Number.

```
SELECT *
FROM student_cgpa
ORDER BY reg_no ASC;
```

OUTPUT:

reg_no	student_name	department	no_of_subs_pass	no_of_subs_fail	cgpa
192111101	Rahul	MECH	30	12	5.14
192312117	Aadhya	ECE	25	3	8.97
192411010	Alekhya	CSE	35	2	9.01
192411122	Yogitha	CSE	31	1	9.01
192411123	Deepthi	CSE	22	2	8.71
192411227	Dimple	CSE	31	0	9.37
192511135	Rahul	CIVIL	37	0	9.72
192512121	Varun	IT	27	1	9.23
192516871	Sadhya	IT	25	5	7.22
192820896	Mohan	EEE	18	9	6.48

10. Display all students in descending order of CGPA.

```
SELECT *
FROM student_cgpa
ORDER BY cgpa DESC;
```

OUTPUT:

reg_no	student_name	department	no_of_subs_pass	no_of_subs_fail	cgpa
192511135	Rahul	CIVIL	37	0	9.72
192411227	Dimple	CSE	31	0	9.37
192512121	Varun	IT	27	1	9.23
192411010	Alekhya	CSE	35	2	9.01
192411122	Yogitha	CSE	31	1	9.01
192312117	Aadhya	ECE	25	3	8.97
192411123	Deepthi	CSE	22	2	8.71
192516871	Sadhya	IT	25	5	7.22
192820896	Mohan	EEE	18	9	6.48
192111101	Rahul	MECH	30	12	5.14

student_cgpa 34

11. Display students in ascending order of Student Name.

```
SELECT *
FROM student_cgpa
ORDER BY student_name ASC;
```

OUTPUT:

reg_no	student_name	department	no_of_subs_pass	no_of_subs_fail	cgpa
192312117	Aadhya	ECE	25	3	8.97
192411010	Alekhya	CSE	35	2	9.01
192411123	Deepthi	CSE	22	2	8.71
192411227	Dimple	CSE	31	0	9.37
192820896	Mohan	EEE	18	9	6.48
192111101	Rahul	MECH	30	12	5.14
192511135	Rahul	CIVIL	37	0	9.72
192516871	Sadhya	IT	25	5	7.22
192512121	Varun	IT	27	1	9.23
192411122	Yogitha	CSE	31	1	9.01

12. Display students in descending order of Number of Passed Subjects.

```
SELECT *
FROM student_cgpa
ORDER BY no_of_subs_pass DESC;
```

OUTPUT:

reg_no	student_name	department	no_of_subs_pass	no_of_subs_fail	cgpa
192511135	Rahul	CIVIL	37	0	9.72
192411010	Alekhya	CSE	35	2	9.01
192411122	Yogitha	CSE	31	1	9.01
192411227	Dimple	CSE	31	0	9.37
192111101	Rahul	MECH	30	12	5.14
192512121	Varun	IT	27	1	9.23
192312117	Aadhya	ECE	25	3	8.97
192516871	Sadhya	IT	25	5	7.22
192411123	Deepthi	CSE	22	2	8.71
192820896	Mohan	EEE	18	9	6.48

13. Display students ordered first by Department and then by CGPA in descending order.

```
SELECT *
FROM student_cgpa
ORDER BY department ASC, cgpa DESC;
```

OUTPUT:

reg_no	student_name	department	no_of_subs_pass	no_of_subs_fail	cgpa
192511135	Rahul	CIVIL	37	0	9.72
192411227	Dimple	CSE	31	0	9.37
192411010	Alekhya	CSE	35	2	9.01
192411122	Yogitha	CSE	31	1	9.01
192411123	Deepthi	CSE	22	2	8.71
192312117	Aadhya	ECE	25	3	8.97
192820896	Mohan	EEE	18	9	6.48
192512121	Varun	IT	27	1	9.23
192516871	Sadhya	IT	25	5	7.22
192111101	Rahul	MECH	30	12	5.14

RESULT

Thus, the records from the student_cgpa table were successfully retrieved and organized using the GROUP BY, HAVING, and ORDER BY clauses in MySQL Workbench.